APPENDIX II

CENTRAL AFRICAN REPUBLIC
DEBT SUSTAINABILITY ANALYSIS USING THE LOW-INCOME COUNTRY FRAMEWORK

According to the analysis based on the joint IMF-World Bank debt sustainability framework for low-income countries,[18] Central African Republic (C.A.R.) has a moderate risk of debt distress.[19] Thanks to debt relief under the enhanced HIPC initiative and MDRI, C.A.R.'s external and public debt burden indicators improve significantly through the projection period. The debt sustainability analysis shows, however, that C.A.R. continues to be vulnerable to certain shocks and could breach the policy-related threshold for the NPV of external debt-to-exports ratio in the most extreme scenario.

I. BACKGROUND

87. **This analysis is based on the forward-looking debt sustainability framework for low-income countries (LIC DSA)**. The LIC DSA uses the reconciled debt database prepared for the completion point HIPC DSA, and incorporates the impact of HIPC, additional bilateral assistance beyond HIPC, and MDRI relief in the baseline scenario. The LIC DSA varies from the HIPC DSA because of three key differences in the methodologies applied: (i) the discount rate in the LIC DSA is fixed at 5 percent, compared to the currency-specific 6-month averages of commercial interest reference rates (CIRR) used in the HIPC DSA; (ii) WEO exchange rate assumptions are used for calculating the present value of debt in the LIC DSA, while an actual exchange rate at end-2008 is used in the HIPC DSA; and (iii) the exports denominator in the LIC DSA is based on the current level of exports of goods and services, rather than the three-year backward-looking average in the HIPC DSA.

88. **This LIC DSA incorporates four key updates relative to the previous one**:[20] (i) debt relief is based on the calculations for the HIPC completion point rather than earlier preliminary estimates; (ii) the baseline scenario assumes lower real GDP growth, larger current account deficit, and worse fiscal balance outcomes in the near term, mainly due to the negative impact of the current global crisis; (iii) real GDP growth is assumed to be lower by

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[18] See "Debt Sustainability in Low-Income Countries: Proposal for an Operational Framework and Policy Implications" (http://www.imf.org/external/np/pdr/sustain/2004/020304.htm) and "Debt Sustainability in Low-Income Countries: Further Considerations on an Operational Framework and Policy Implications" (http://www.imf.org/external/np/pdr/sustain/2004/091004.htm)

[19] The LIC DSA compares the evolution over the projection period of debt-burden indicators against policy-dependent indicative thresholds, using the three-year average of the World Bank's Country Policy and Institutional Assessment (CPIA). C.A.R.'s policies and institutions, as measured by the World Bank's (CPIA), are classified as a "weak performer."

[20] The previous joint LIC DSA was published in December 2008 (IMF Country Report No. 09/43; IDA Report No. 47247).

about 0.7 percentage point over the period; and (iv) the historical data for real GDP growth were recently revised by the authorities following technical updates, which affect the level of nominal GDP in 2009 and thereafter.

89. **Total public debt including domestic arrears of C.A.R. is estimated at 79.4 percent of GDP at the end of 2008**. External public and publicly guaranteed debt amounts to 56.8 percent of GDP, of which multilateral creditors account for more than half and official bilateral creditors for about one-third. Domestic public debt (including budgetary arrears and domestic debt of public enterprises) amounts to 23 percent of GDP. It consists of outstanding credits to the government from domestic commercial banks (10 percent), government debt with the Bank for Central African States (BEAC, 30 percent), budgetary arrears (52 percent), public enterprise debt (4 percent), and nonbanks (5 percent).

II. Underlying DSA Assumptions

90. **The near-term macroeconomic outlook is influenced by the adverse impact of the international financial crisis**. Despite favorable developments for the import price of oil, the sudden drop of timber and diamonds exports implies lower GDP growth and a worse external current account balance. Along with lower revenue projected for 2009, the additional spending necessary to consolidate the peace process and finance the 2010 elections moves the domestic primary balance into negative territory in 2009; however, the primary balance including grants remains in surplus.

91. **Over the medium and long term, C.A.R. is presumed to achieve a steady-state growth path, supported by political and social stability**. A sustained improvement in business confidence and higher investment should underpin these growth projections (Box 1). Growing exports of a diversified range of primary goods, including gold and uranium, are expected to lift real GDP growth to 5.5 percent in the medium term. Stronger exports will help improve the external current account deficit over time to around 4 percent of GDP, with financing primarily provided by private investment, highly concessional project loans, and the regional market for government securities expected to be launched in 2010. In order to preserve debt sustainability, fiscal policy would remain prudent. The authorities' fiscal anchor is the domestic primary balance, which would be in surplus until 2019. The primary balance including grants would be higher than the primary balance needed to stabilize the debt-to-GDP ratio for the projection period; thus, the debt-to-GDP ratio could decline by some 10 percentage points to 20 percent of GDP by 2029.

92. **The risks to C.A.R.'s macroeconomic outlook, however, remain significant**. Potentially, political uncertainty and a worsening social and security situation could hamper donor support and investor confidence. Exogenous shocks, including a prolonged impact of the global economic crisis on exports, could lead to slower growth, lower revenue, and a deterioration in debt indicators. Moreover, insufficient investments in infrastructure, such as

roads, could cause delays in private sector investment projects that are needed to boost the C.A.R. economy and reduce poverty.

Box 1. Baseline Macroeconomic Assumptions

Real GDP growth: Average annual real GDP growth for 2009–29 is projected at 4.4 percent, which is predicated on sustained security and political stability, an improvement in the country's institutional and administrative capacity, and appropriate macroeconomic policies. This environment should encourage an increase in private investment, especially in forestry, mining, and telecommunications. Public investment in infrastructure would help revive agriculture, which dominates economic activity. With these assumptions, the projected growth rate is significantly higher than the historical average, which was characterized by conflicts and civil strife.

Inflation: After unexpectedly high inflation in 2008, the GDP deflator is projected to increase by 2½ percent on average for 2009–29; this assumes that inflation will moderate promptly from the current level. The projected inflation rate is in line with the Central African Economic and Monetary Community (CEMAC)'s convergence criterion of 3 percent (defined by the CPI).

Current account balance: The current account deficit (including grants) is projected to average 7 percent for 2009–29. The trade balance is projected to improve over time, driven by stronger export performance as a result of structural reforms and infrastructure investments that will enhance competitiveness and diversification of the export base; the deficits in the services balance would remain large. The current account deficits would be financed primarily by official development assistance (project loans), foreign direct investment, and regional capital inflows from the future regional government securities market. A major gold mining project is assumed to start in 2011. The new forestry and mining codes should prepare the ground for sustained FDI inflows in these sectors.

Government balance: The domestic primary surplus would be in surplus until 2019 and then decline to zero between 2020–29. The overall fiscal deficit (including grants) is projected to average about 1.2 percent of GDP for 2009–29; with the primary balance consistently higher than the one needed to stabilize the debt-to-GDP ratio. Domestic revenue is projected to rise from 10½ percent of GDP in 2009 to some 14 percent at end-2029, mainly as a result of steady tax and customs administration improvements and tax policy reform to render the tax code more growth compatible. Expenditures are projected to rise from about 17 percent of GDP in 2009 to about 20 percent in 2029 with most of the increases concentrated in infrastructure investments.

External assistance: Total grants and loans are assumed to contribute about 5.4 percent of nominal GDP in the long run. Grants are assumed to account for about 80 percent of total external assistance, with the grant element of new external loans averaging 52 percent for the period.

Domestic borrowing: It is assumed that in 2010, the government will start accessing the securities markets that are being developed in the CEMAC region. This will allow it to reduce financing costs, improve liquidity managements, and repay domestic arrears. Given the continued prudent fiscal policy stance, domestic debt is expected to decline during the projection period.

Real interest rate on domestic currency debt: The average real interest rate on domestic currency debt (including bonds from the regional markets) should converge to about 4.8 percent in the long run.

III. External Debt Sustainability analysis

93. **Following HIPC and MDRI relief, all debt indicators are expected to remain below the relevant thresholds throughout the projection period in the baseline scenario (Text Table 1, Figure 1, and Table 1a)**. The debt service-to-export ratio and the debt service-to-revenue ratio increase temporarily over the medium term due to debt service payments on new debts following the standard grace periods, but the ratios would stay well below the thresholds that define an excessive debt burden. All other debt indicators remain well below the thresholds for the projection period and show stable downward trends.

Text Table 1. Central African Republic: Policy-Based Thresholds and External Debt Burden Indicators

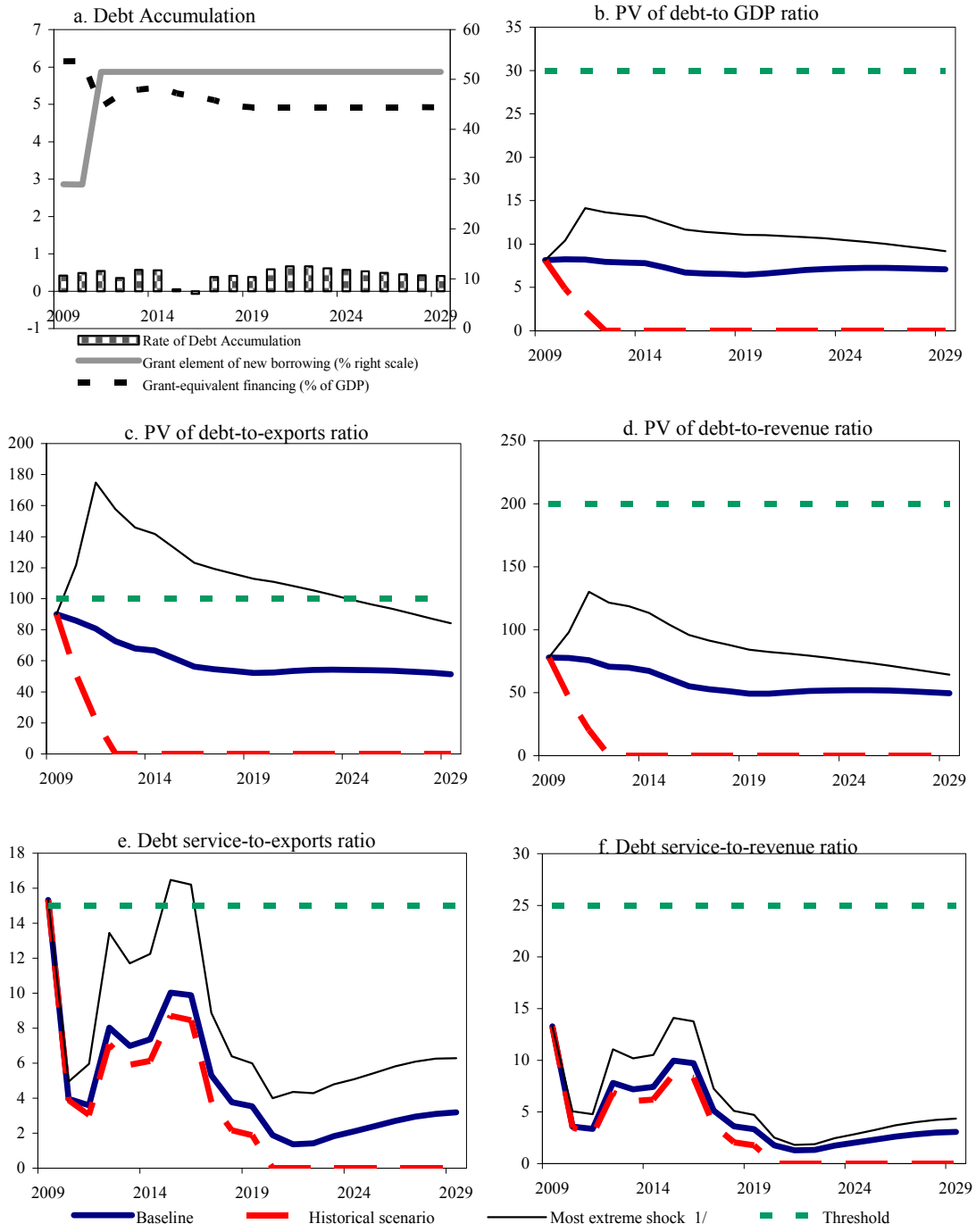
	Thresholds [1]	Baseline Scenario Ratios		
		2008 Est.	2009−29[2] Proj.	2009−29 Peak
NPV of external debt in percent of:				
GDP	30	34.9	7.3	8.2
Exports	100	323.5	60.7	90.0
Revenue	200	336.4	58.0	78.0
External debt service in percent of:				
Exports	15	27.5	4.8	15.3
Revenue	25	28.6	4.6	13.3

Sources: C.A.R. authorities; and IMF and World Bank staffs estimates.

[1] Policy-dependent thresholds used in the joint IMF-WB LIC DSA framework is for a weak policy performer. C.A.R. received an average rating of 2.43 in 2005−07 in the World Bank's Country Policy and Institutional Assessment (CPIA), which qualifies it as a weak policy performer.
[2] Simple average.

94. **The historical scenario appears much more favorable than the baseline scenario because of severely compressed imports during C.A.R.'s history of conflict (Figure 1 and Table 1b)**. Thus, the historical scenario is based on the average noninterest current account deficit of only 2.2 percent of GDP, compared to the projected deficit of 7 percent of GDP under the baseline scenario; such scenario would be unlikely given recent trends and detrimental to economic growth and integration of C.A.R. into the world economy.

Figure 1. Central African Republic: Indicators of Public and Publicly Guaranteed External Debt under Alternatives Scenarios, 2009–29 1/



Source: C.A.R. authorities; and IMF and World Bank staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio in 2019. In figure b. it corresponds to a Combination shock; in c. to a Combination shock; in d. to a Combination shock; in e. to a Exports shock and in picture f. to a One-time depreciation shock

95. **The bound tests underscore potential risks, especially regarding lower export growth**. The most extreme stress test, which assumes a combination of shocks but mostly affected by lower export value growth in the C.A.R.'s case, would raise the NPV of the debt-to-exports ratio above the threshold for a prolonged period (Figure 1c and Table 1b).[21] The debt service-to-exports ratio would also exceed the threshold during 2015–16 under the low export growth scenario (Figure 1e and Table 1b). The deterioration of debt indicators under these scenarios underline that continuous efforts are necessary to develop C.A.R.'s exports, including improving the business climate for foreign investments, reducing transport costs, and improving productivity.

IV. Public Debt Sustainability analysis

96. **HIPC and MDRI debt relief will improve public debt indicators significantly since public external debt accounts for almost half of public debt stock (Figure 2 and Table 2a)**. With an assumption of lower program grants than in the recent past (0.7 percent of GDP over the long term), and with well-managed financing, all debt indicators are expected to decline gradually. It is assumed that the government would continue its prudent fiscal policy by maintaining a domestic primary balance surplus in the medium term and balance over the long term; mobilizing a higher domestic revenue-to-GDP ratio; and securing relatively low-cost financing such as highly concessional loans. Accessing the regional government securities markets starting in 2010 would help eliminate expensive credits from commercial banks over the next several years and clear domestic arrears. Under these assumptions, the NPV of the public debt-to-GDP ratio would decline from 28 percent of GDP in 2009 to 17 percent of GDP in 2029, while the NPV of public debt-to-revenue ratio would fall from 174 percent to 90 percent.

97. **All debt indicators would deteriorate rapidly if real GDP growth were slower or debt-creating flows were to increase (Figure 2 and Table 2b)**. Lower GDP growth represents the most extreme scenario: in this case, both the NPV of the debt-to-GDP ratio and the NPV of the debt-to-revenue ratio as well as the debt service-to-revenue ratio would rise overtime.

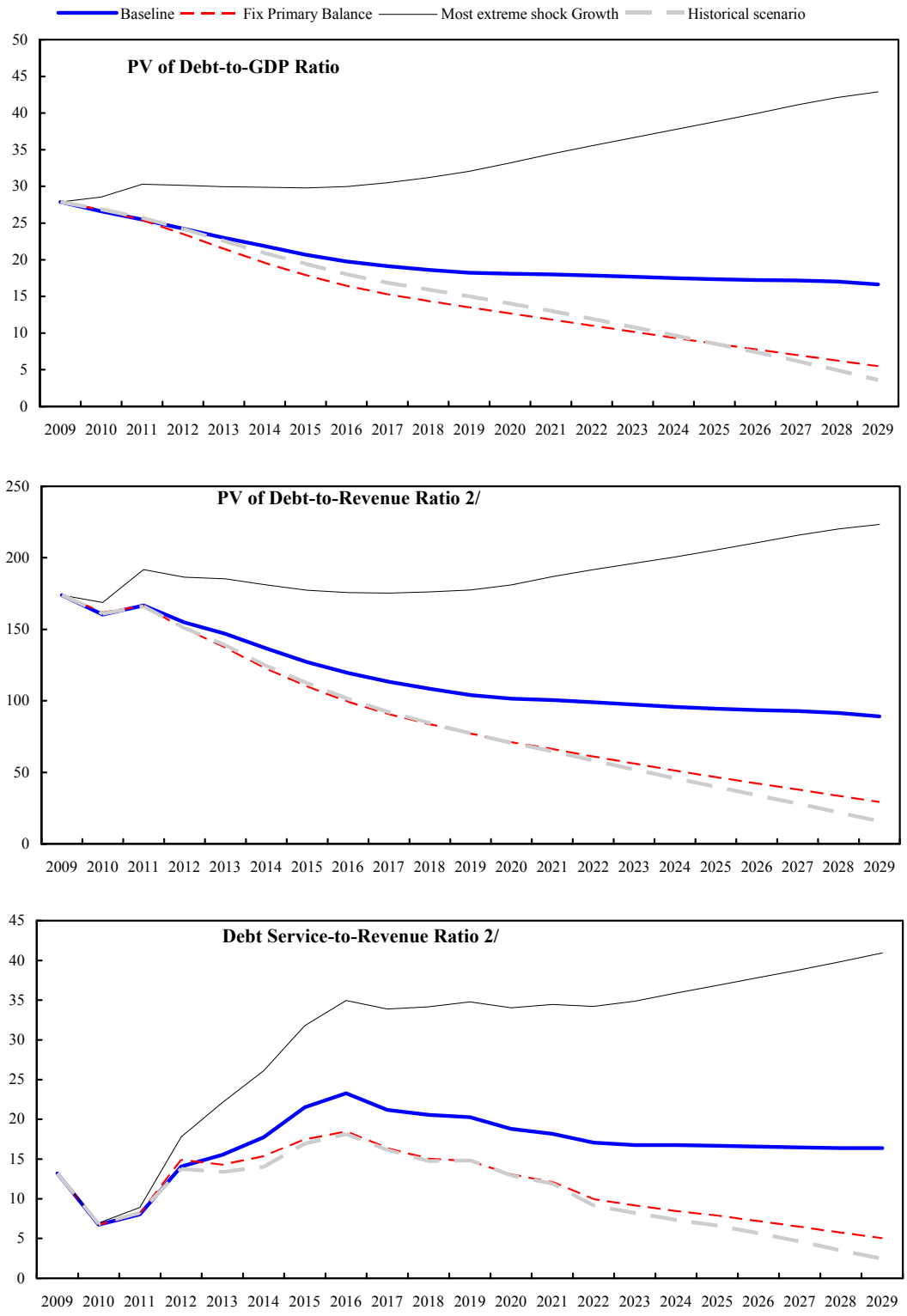
V. Debt Distress Qualification and Conclusions

98. **C.A.R. qualifies as having a "moderate risk of debt distress"**. All debt indicators improve dramatically following the HIPC completion point and debt relief under the MDRI,

[21] As shown on the Table 1b, the combination of shocks consists of (i) real GDP growth at historical average minus one-half standard deviation in 2010–11; (ii) export value growth at historical average minus one-half standard deviation in 2010–11; (iii) US dollar GDP deflator at historical average minus one-half standard deviation in 2010–11; and (iv) net non-debt creating flows at historical average minus one standard deviation in 2010–11.

but its overall debt position could still be vulnerable to a variety of shocks. The external debt indicators are particularly sensitive to export growth, indicating that diversifying the export base is essential for preserving external debt sustainability. The public debt indicators are vulnerable to slower GDP growth and an increase in debt-creating flows, confirming that C.A.R. needs to pursue prudent fiscal policies over the medium-term and to consolidate the basis for growth by fostering domestic security, maintaining political stability, and improving the country's institutional and administrative capacity.

Figure 2. Central African Republic: Indicators of Public Debt Under Alternative Scenarios, 2009–29 1/



Sources: C.A.R. authorities; and IMF and World Bank staff estimates and projections.
1/ The most extreme stress test is the test that yields the highest ratio in 2019.
2/ Revenues are defined inclusive of grants.

Table 1a. Central African Republic: External Debt Sustainability Framework, Baseline Scenario, 2006–29 1/
(In percent of GDP, unless otherwise indicated)

	Actual			Historical 0 Average 0	Standard Deviation	Projections									
	2006	2007	2008			2009	2010	2011	2012	2013	2014	2009-2014 Average	2019	2029	2015-2029 Average
External debt (nominal) 1/	**70.1**	**54.6**	**56.8**			**9.8**	**9.8**	**10.0**	**10.3**	**10.9**	**11.5**		**11.1**	**11.7**	
o/w public and publicly guaranteed (PPG)	70.1	54.6	56.8			9.8	9.8	10.0	10.3	10.9	11.5		11.1	11.7	
Change in external debt	-11.6	-15.4	2.2			-47.0	0.0	0.2	0.3	0.6	0.6		-0.1	-0.1	
Identified net debt-creating flows	-6.0	-6.4	-4.6			6.3	6.7	5.7	4.8	4.1	3.8		2.6	0.7	
Non-interest current account deficit	**2.4**	**5.7**	**8.8**	**2.9**	**3.2**	**9.0**	**9.6**	**9.6**	**9.3**	**8.6**	**8.4**		**7.0**	**4.0**	6.1
Deficit in balance of goods and services	7.7	9.4	12.6			12.5	13.0	12.5	12.1	11.4	11.1		9.6	6.7	
Exports	14.2	14.1	10.8			9.1	9.6	10.2	10.9	11.6	11.7		12.4	13.8	
Imports	21.9	23.5	23.4			21.6	22.6	22.7	23.0	23.0	22.8		22.0	20.5	
Net current transfers (negative = inflow)	-5.5	-3.8	-3.8	-4.0	1.1	-4.1	-3.5	-2.9	-2.9	-2.9	-2.9		-2.8	-2.7	-2.8
o/w official	-5.3	-3.5	-3.3			-3.8	-3.2	-2.7	-2.7	-2.7	-2.7		-2.8	-2.8	
Other current account flows (negative = net inflow)	0.2	0.1	0.1			0.6	0.1	0.1	0.1	0.1	0.1		0.1	0.1	
Net FDI (negative = inflow)	**-2.3**	**-3.3**	**-6.2**	**-2.0**	**1.8**	**-1.8**	**-2.8**	**-3.8**	**-4.2**	**-4.2**	**-4.2**		**-4.0**	**-3.0**	-3.7
Endogenous debt dynamics 2/	**-6.0**	**-8.8**	**-7.2**			**-0.9**	**-0.1**	**-0.2**	**-0.3**	**-0.4**	**-0.4**		**-0.4**	**-0.4**	
Contribution from nominal interest rate	0.6	0.4	1.0			0.6	0.2	0.2	0.2	0.2	0.1		0.1	0.1	
Contribution from real GDP growth	-2.9	-2.2	-1.0			-1.4	-0.3	-0.4	-0.5	-0.5	-0.6		-0.5	-0.5	
Contribution from price and exchange rate changes	-3.7	-7.0	-7.1			…	…	…	…	…	…		…	…	
Residual 3/4/	**-5.6**	**-9.0**	**6.7**			**-53.3**	**-6.7**	**-5.5**	**-4.5**	**-3.5**	**-3.2**		**-2.7**	**-0.7**	
o/w exceptional financing	5.3	-0.1	-1.9			-2.2	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
PV of external debt 5/	...	...	34.9			8.2	8.2	8.2	8.0	7.9	7.8		6.5	7.1	
In percent of exports	...	...	323.5			90.0	85.8	80.6	72.7	67.9	66.6		52.2	51.4	
PV of PPG external debt	**...**	**...**	**34.9**			**8.2**	**8.2**	**8.2**	**8.0**	**7.9**	**7.8**		**6.5**	**7.1**	
In percent of exports	**...**	**...**	**323.5**			**90.0**	**85.8**	**80.6**	**72.7**	**67.9**	**66.6**		**52.2**	**51.4**	
In percent of government revenues	**...**	**...**	**336.4**			**78.0**	**77.5**	**75.7**	**70.6**	**69.8**	**67.3**		**49.2**	**49.6**	
Debt service-to-exports ratio (in percent)	**18.6**	**14.1**	**27.5**			**15.3**	**4.0**	**3.6**	**8.0**	**7.0**	**7.4**		**3.5**	**3.2**	
PPG debt service-to-exports ratio (in percent)	**18.6**	**14.1**	**27.5**			**15.3**	**4.0**	**3.6**	**8.0**	**7.0**	**7.4**		**3.5**	**3.2**	
PPG debt service-to-revenue ratio (in percent)	**27.8**	**19.4**	**28.6**			**13.3**	**3.6**	**3.4**	**7.8**	**7.2**	**7.4**		**3.3**	**3.1**	
Total gross financing need (Billions of U.S. dollars)	0.0	0.1	0.1			0.2	0.1	0.1	0.1	0.1	0.1		0.1	0.1	
Non-interest current account deficit that stabilizes debt ratio	14.0	21.2	9.7			56.0	9.6	9.4	9.0	8.1	7.8		7.0	4.1	
Key macroeconomic assumptions															
Real GDP growth (in percent)	3.8	3.7	2.2	1.1	3.2	2.4	3.1	4.0	5.0	5.5	5.5	4.3	4.5	4.3	4.5
GDP deflator in US dollar terms (change in percent)	4.8	11.1	15.0	6.1	10.1	-7.5	2.0	2.5	2.7	2.6	2.3	0.8	2.4	2.4	2.4
Effective interest rate (percent) 6/	0.8	0.7	2.1	1.2	0.5	1.0	2.5	2.4	2.0	1.7	1.4	1.8	0.9	0.9	1.0
Growth of exports of G&S (US dollar terms, in percent)	21.3	14.6	-10.0	0.9	12.0	-20.5	11.2	13.4	15.7	14.6	9.3	7.3	8.3	7.5	8.1
Growth of imports of G&S (US dollar terms, in percent)	14.6	23.6	17.0	5.9	14.4	-12.4	9.8	7.1	9.4	8.2	7.2	4.9	6.3	5.9	6.2
Grant element of new public sector borrowing (in percent)	...	...	...	...	...	28.9	28.9	51.5	51.5	51.5	51.5	44.0	51.5	51.5	51.5
Government revenues (excluding grants, in percent of GDP)	9.5	10.3	10.4			10.5	10.6	10.9	11.3	11.3	11.6		13.1	14.3	13.4
Aid flows (in Billions of US dollars) 7/	0.2	0.1	0.1			0.1	0.1	0.1	0.1	0.2	0.2		0.2	0.4	
o/w Grants	0.2	0.1	0.1			0.1	0.1	0.1	0.1	0.1	0.1		0.2	0.3	
o/w Concessional loans	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.1		0.0	0.1	
Grant-equivalent financing (in percent of GDP) 8/	...	...	...			6.2	6.2	4.9	5.3	5.4	5.4		4.9	4.9	5.0
Grant-equivalent financing (in percent of external financing) 8/	...	...	...			81.1	93.1	91.1	86.5	84.8	84.3		91.0	91.0	90.2
Memorandum items:															
Nominal GDP (Billions of US dollars)	1.5	1.7	2.0			1.9	2.0	2.1	2.3	2.5	2.7		3.8	7.4	
Nominal dollar GDP growth	8.8	15.3	17.5			-5.3	5.1	6.6	7.8	8.3	8.0	**5.1**	7.0	6.8	7.0
PV of PPG external debt (in Billions of US dollars)			0.1			0.2	0.2	0.2	0.2	0.2	0.2		0.2	0.5	
(PVt-PVt-1)/GDPt-1 (in percent)						0.4	0.5	0.5	0.4	0.6	0.6	0.5	0.4	0.4	0.4

Source: C.A.R. authorities; IMF and World Bank staff estimates and projections.

1/ Includes both public and private sector external debt.
2/ Derived as [r - g - r(1+g)]/(1+g+r+gr) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and r = growth rate of GDP deflator in U.S. dollar terms.
3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.
4/ The figure for Year 2009 reflects debt relief under the HIPC initiative and MDRI.
5/ Assumes that PV of private sector debt is equivalent to its face value.
6/ Current-year interest payments divided by previous period debt stock.
7/ Defined as grants, concessional loans, and debt relief.
8/ Grant-equivalent financing includes grants provided directly to the government and through new borrowing (difference between the face value and the PV of new debt).

Table 1b. Central African Republic: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2009–29
(In percent)

											Projections										
	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	**2019**	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029
PV of debt-to GDP ratio																					
Baseline	8	8	8	8	8	8	7	7	7	7	**6**	7	7	7	7	7	7	7	7	7	7
A. Alternative Scenarios																					
A1. Key variables at their historical averages in 2009–29 1/	8	5	2	0	0	0	0	0	0	0	**0**	0	0	0	0	0	0	0	0	0	0
A2. New public sector loans on less favorable terms in 2009–29 2	8	8	9	9	9	10	10	9	10	10	**10**	10	10	11	11	11	11	12	12	12	12
B. Bound Tests																					
B1. Real GDP growth at historical average minus one standard deviation in 2010–11	8	9	9	9	9	9	8	8	7	7	**7**	7	8	8	8	8	8	8	8	8	8
B2. Export value growth at historical average minus one standard deviation in 2010–11 3/	8	9	11	11	11	10	10	9	9	9	**9**	9	9	9	9	9	9	8	8	8	8
B3. US dollar GDP deflator at historical average minus one standard deviation in 2010–11	8	9	9	9	9	9	8	8	7	7	**7**	7	8	8	8	8	8	8	8	8	8
B4. Net non-debt creating flows at historical average minus one standard deviation in 2010–11 4/	8	10	12	11	11	11	10	10	9	9	**9**	9	9	9	9	9	9	9	8	8	8
B5. Combination of B1-B4 using one-half standard deviation shocks	8	10	14	14	13	13	12	12	11	11	**11**	11	11	11	11	10	10	10	10	9	9
B6. One-time 30 percent nominal depreciation relative to the baseline in 2010 5/	8	12	12	11	11	11	10	10	9	9	**9**	9	10	10	10	10	10	10	10	10	10
PV of debt-to-exports ratio																					
Baseline	90	86	81	73	68	67	62	56	55	53	**52**	53	53	54	54	54	54	54	53	52	51
A. Alternative Scenarios																					
A1. Key variables at their historical averages in 2009–29 1/	90	51	22	0	0	0	0	0	0	0	**0**	0	0	0	0	0	0	0	0	0	0
A2. New public sector loans on less favorable terms in 2009–29 2	90	87	85	81	80	83	81	78	79	79	**79**	80	82	83	84	85	85	86	86	86	86
B. Bound Tests																					
B1. Real GDP growth at historical average minus one standard deviation in 2010–11	90	86	81	73	68	67	62	56	55	53	**52**	53	53	54	54	54	54	54	53	52	51
B2. Export value growth at historical average minus one standard deviation in 2010–11 3/	90	120	174	157	146	142	132	122	119	116	**113**	111	110	108	106	104	101	99	97	94	91
B3. US dollar GDP deflator at historical average minus one standard deviation in 2010–11	90	86	81	73	68	67	62	56	55	53	**52**	53	53	54	54	54	54	54	53	52	51
B4. Net non-debt creating flows at historical average minus one standard deviation in 2010–11 4/	90	103	115	103	96	93	87	81	78	76	**74**	73	72	70	69	67	65	64	62	60	58
B5. Combination of B1-B4 using one-half standard deviation shocks	90	122	175	158	146	142	132	123	119	116	**113**	111	108	106	102	99	96	93	90	87	84
B6. One-time 30 percent nominal depreciation relative to the baseline in 2010 5/	90	86	81	73	68	67	62	56	55	53	**52**	53	53	54	54	54	54	54	53	52	51
PV of debt-to-revenue ratio																					
Baseline	78	78	76	71	70	67	61	55	53	51	**49**	49	50	51	52	52	52	52	51	50	50
A. Alternative Scenarios																					
A1. Key variables at their historical averages in 2009–29 1/	78	46	20	0	0	0	0	0	0	0	**0**	0	0	0	0	0	0	0	0	0	0
A2. New public sector loans on less favorable terms in 2009–29 2	78	78	80	79	82	83	80	77	76	76	**74**	75	77	79	80	81	82	83	83	83	83
B. Bound Tests																					
B1. Real GDP growth at historical average minus one standard deviation in 2010–11	78	82	85	79	78	75	68	62	59	57	**55**	55	56	57	58	58	58	58	57	56	55
B2. Export value growth at historical average minus one standard deviation in 2010–11 3/	78	87	103	96	94	90	82	75	72	69	**67**	65	65	65	64	62	61	60	58	57	55
B3. US dollar GDP deflator at historical average minus one standard deviation in 2010–11	78	82	86	80	79	76	69	63	60	58	**56**	56	57	58	59	59	59	59	58	57	56
B4. Net non-debt creating flows at historical average minus one standard deviation in 2010–11 4/	78	93	108	101	99	94	86	79	76	73	**70**	68	68	67	66	64	63	61	60	58	56
B5. Combination of B1-B4 using one-half standard deviation shocks	78	98	130	121	119	113	104	96	91	88	**84**	82	81	79	78	76	74	71	69	67	64
B6. One-time 30 percent nominal depreciation relative to the baseline in 2010 5/	78	110	107	100	99	95	87	78	75	72	**70**	70	72	73	73	74	74	73	72	71	70

Table 1b. Central African Republic: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2009–29 (concluded)
(In percent)

											Projections										
	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	**2019**	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029
Debt service-to-exports ratio																					
Baseline	15	4	4	8	7	7	10	10	5	4	**4**	2	1	1	2	2	2	3	3	3	3
A. Alternative Scenarios																					
A1. Key variables at their historical averages in 2009–29 1/	15	4	3	7	6	6	9	8	4	2	**2**	0	0	0	0	0	0	0	0	0	0
A2. New public sector loans on less favorable terms in 2009–29 2	15	4	4	8	7	8	11	10	6	4	**4**	3	3	3	4	4	4	4	4	5	5
B. Bound Tests																					
B1. Real GDP growth at historical average minus one standard deviation in 2010–11	15	4	4	8	7	7	10	10	5	4	**4**	2	1	1	2	2	2	3	3	3	3
B2. Export value growth at historical average minus one standard deviation in 2010–11 3/	15	5	6	13	12	12	16	16	9	6	**6**	4	4	4	5	5	5	6	6	6	6
B3. US dollar GDP deflator at historical average minus one standard deviation in 2010–11	15	4	4	8	7	7	10	10	5	4	**4**	2	1	1	2	2	2	3	3	3	3
B4. Net non-debt creating flows at historical average minus one standard deviation in 2010–11 4/	15	4	4	9	7	8	10	10	6	4	**4**	3	3	3	3	3	4	4	4	4	4
B5. Combination of B1-B4 using one-half standard deviation shocks	15	5	5	12	11	11	15	14	8	6	**5**	4	5	5	5	5	6	6	6	6	6
B6. One-time 30 percent nominal depreciation relative to the baseline in 2010 5/	15	4	4	8	7	7	10	10	5	4	**4**	2	1	1	2	2	2	3	3	3	3
Debt service-to-revenue ratio																					
Baseline	13	4	3	8	7	7	10	10	5	4	**3**	2	1	1	2	2	2	3	3	3	3
A. Alternative Scenarios																					
A1. Key variables at their historical averages in 2009–29 1/	13	3	3	7	6	6	9	8	4	2	**2**	0	0	0	0	0	0	0	0	0	0
A2. New public sector loans on less favorable terms in 2009–29 2	13	4	3	8	8	8	11	10	6	4	**4**	3	3	3	3	4	4	4	4	4	4
B. Bound Tests																					
B1. Real GDP growth at historical average minus one standard deviation in 2010–11	13	4	4	9	8	8	11	11	6	4	**4**	2	1	1	2	2	3	3	3	3	3
B2. Export value growth at historical average minus one standard deviation in 2010–11 3/	13	4	4	8	8	8	10	10	5	4	**4**	2	3	3	3	3	3	4	4	4	4
B3. US dollar GDP deflator at historical average minus one standard deviation in 2010–11	13	4	4	9	8	8	11	11	6	4	**4**	2	1	2	2	2	3	3	3	3	3
B4. Net non-debt creating flows at historical average minus one standard deviation in 2010–11 4/	13	4	4	8	8	8	10	10	5	4	**4**	3	3	3	3	3	3	4	4	4	4
B5. Combination of B1-B4 using one-half standard deviation shocks	13	4	4	9	9	9	12	11	6	4	**4**	3	4	4	4	4	4	4	5	5	5
B6. One-time 30 percent nominal depreciation relative to the baseline in 2010 5/	13	5	5	11	10	11	14	14	7	5	**5**	3	2	2	2	3	3	4	4	4	4
Memorandum item:																					
Grant element assumed on residual financing (i.e., financing required above baseline) 6/	48	48	48	48	48	48	48	48	48	48	**48**	48	48	48	48	48	48	48	48	48	48

Source: IMF and World Bank staff estimates and projections.

1/ Variables include real GDP growth, growth of GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.
2/ Assumes that the interest rate on new borrowing is by 2 percentage points higher than in the baseline., while grace and maturity periods are the same as in the baseline.
3/ Exports values are assumed to remain permanently at the lower level, but the current account as a share of GDP is assumed to return to its baseline level after the shock (implicitly assuming an offsetting adjustment in import levels).
4/ Includes official and private transfers and FDI.
5/ Depreciation is defined as percentage decline in dollar/local currency rate, such that it never exceeds 100 percent.
6/ Applies to all stress scenarios except for A2 (less favorable financing) in which the terms on all new financing are as specified in footnote 2.

Table 2a. Central African Republic: Public Sector Debt Sustainability Framework, Baseline Scenario, 2006–29
(In percent of GDP, unless otherwise indicated)

	Actual					Estimate	Projections								
	2006	2007	2008	Average	Standard Deviation	2009	2010	2011	2012	2013	2014	2009-14 Average	2019	2029	2015-29 Average
Public sector debt 1/	93.9	79.1	79.4			29.5	28.1	27.3	26.6	26.0	25.5		22.9	21.2	
o/w foreign-currency denominated	70.1	54.6	56.8			9.8	9.8	10.0	10.3	10.9	11.5		11.1	11.7	
Change in public sector debt	-12.1	-14.8	0.3			-49.9	-1.4	-0.9	-0.7	-0.6	-0.5		-0.4	-0.4	
Identified debt-creating flows	-23.3	-12.4	-2.9			-1.7	-1.2	-0.8	-0.7	-0.6	-0.5		-0.4	-0.4	
Primary deficit	-9.9	-2.5	-1.5	-0.9	3.6	-0.4	-0.7	-0.1	0.3	0.4	0.5	0.0	0.2	0.2	0.3
Revenue and grants	22.9	14.4	15.1			16.0	16.6	15.3	15.6	15.7	16.0		17.5	18.7	
of which: grants	13.4	4.1	4.7			5.6	6.0	4.4	4.4	4.4	4.4		4.4	4.4	
Primary (noninterest) expenditure	13.0	11.9	13.6			15.6	15.9	15.2	15.9	16.1	16.4		17.7	18.9	
Automatic debt dynamics	-13.4	-9.9	-1.4			-1.3	-0.5	-0.7	-1.0	-1.0	-1.0		-0.5	-0.6	
Contribution from interest rate/growth differential	-6.3	-4.3	-2.5			-1.9	-0.4	-0.6	-0.9	-1.0	-0.9		-0.5	-0.5	
of which: contribution from average real interest rate	-2.4	-1.0	-0.8			0.0	0.5	0.5	0.4	0.4	0.4		0.5	0.4	
of which: contribution from real GDP growth	-3.9	-3.4	-1.7			-1.9	-0.9	-1.1	-1.3	-1.4	-1.4		-1.0	-0.9	
Contribution from real exchange rate depreciation	-7.1	-5.6	1.1			0.6	-0.2	-0.1	-0.1	-0.1	0.0		...	...	
Other identified debt-creating flows	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Privatization receipts (negative)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Recognition of implicit or contingent liabilities	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Debt relief (HIPC and other)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Other (specify, e.g. bank recapitalization)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Residual, including asset changes	11.3	-2.4	3.3			-48.2	-0.1	0.0	0.0	0.0	0.0		0.0	0.0	
Other Sustainability Indicators															
PV of public sector debt	23.9	24.5	30.5			27.9	26.6	25.5	24.2	23.0	21.9		18.2	16.6	
o/w foreign-currency denominated	0.0	0.0	7.9			8.2	8.2	8.2	7.9	7.9	7.8		6.5	7.1	
o/w external	...	...	7.9			8.2	8.2	8.2	7.9	7.9	7.8		6.5	7.1	
PV of contingent liabilities (not included in public sector debt)	...	...	...			...	...	...	...	...	...		...	...	
Gross financing need 2/	-6.9	0.2	2.3			1.7	0.5	1.1	2.5	2.9	3.3		3.7	3.3	
PV of public sector debt-to-revenue and grants ratio (in percent)	104.3	170.2	201.8			173.9	160.2	166.7	154.9	146.9	136.7		103.9	89.1	
PV of public sector debt-to-revenue ratio (in percent)	250.6	238.2	294.0			266.5	250.4	234.6	215.2	203.7	188.1		138.7	116.6	
o/w external 3/	…	…	76.1			78.1	77.5	75.6	70.6	69.8	67.3		49.2	49.6	
Debt service-to-revenue and grants ratio (in percent) 4/	13.1	18.8	25.3			13.2	6.7	8.0	14.1	15.5	17.7		20.3	16.4	
Debt service-to-revenue ratio (in percent) 4/	31.5	26.3	36.9			20.2	10.5	11.3	19.5	21.5	24.4		27.1	21.4	
Primary deficit that stabilizes the debt-to-GDP ratio	2.1	12.3	-1.8			49.5	0.7	0.8	1.0	1.0	1.0		0.6	0.6	
Key macroeconomic and fiscal assumptions															
Real GDP growth (in percent)	3.8	3.7	2.2	1.1	3.2	2.4	3.1	4.0	5.0	5.5	5.5	4.3	4.5	4.3	4.5
Average nominal interest rate on forex debt (in percent)	0.8	0.7	2.1	1.2	0.5	1.0	2.5	2.4	2.0	1.7	1.4	1.8	0.9	0.9	1.0
Average real interest rate on domestic debt (in percent)	-2.5	1.3	-3.0	-1.5	1.9	-0.5	1.6	2.1	2.4	2.9	3.3	2.0	5.2	5.0	4.8
Real exchange rate depreciation (in percent, + indicates depreciation)	-9.2	-8.4	2.0	-5.4	8.8	1.0	...	...	...	...	...	...	...	...	...
Inflation rate (GDP deflator, in percent)	4.2	1.8	7.1	2.8	2.1	3.9	2.3	2.4	2.4	2.4	2.4	2.6	2.4	2.4	2.4
Growth of real primary spending (deflated by GDP deflator, in percent)	-0.2	0.0	0.2	0.0	0.2	0.2	0.1	0.0	0.1	0.1	0.1	0.1	0.1	0.0	0.1
Grant element of new external borrowing (in percent)	...	...	...	…	…	28.9	28.9	51.5	51.5	51.5	51.5	44.0	51.5	51.5	...

Sources: C.A.R. authorities; and IMF and World Bank staff estimates and projections.

1/ Includes public and publicly-guaranteed external debt, domestic public debt, budgetary arrears of the central government, and domestic debt of state-owned enterprises.

2/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period.

3/ Revenues excluding grants.

4/ Debt service is defined as the sum of interest and amortization of medium and long-term debt.

Table 2b. Central African Republic: Sensitivity Analysis for Key Indicators of Public Debt 2009–29

	Projections							
	2009	2010	2011	2012	2013	2014	2019	2029
PV of Debt-to-GDP Ratio								
Baseline	28	27	25	24	23	22	18	17
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	28	27	26	24	23	21	15	4
A2. Primary balance is unchanged from 2009	28	27	25	24	21	20	14	5
A3. Permanently lower GDP growth 1/	28	27	26	25	24	24	24	37
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2010–11	28	29	30	30	30	30	32	43
B2. Primary balance is at historical average minus one standard deviations in 2010–11	28	30	31	30	29	27	23	21
B3. Combination of B1-B2 using one half standard deviation shocks	28	29	30	30	29	29	29	36
B4. One-time 30 percent real depreciation in 2010	28	30	29	27	25	24	19	17
B5. 10 percent of GDP increase in other debt-creating flows in 2010	28	36	35	33	32	31	26	23
PV of Debt-to-Revenue Ratio 2/								
Baseline	174	160	167	155	147	137	104	89
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	174	161	166	151	139	124	77	16
A2. Primary balance is unchanged from 2009	174	162	166	150	137	122	77	29
A3. Permanently lower GDP growth 1/	174	161	170	160	154	147	133	192
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2010–11	174	169	192	187	185	181	177	223
B2. Primary balance is at historical average minus one standard deviations in 2010–11	174	180	206	192	183	171	131	110
B3. Combination of B1-B2 using one half standard deviation shocks	174	173	193	185	182	175	161	188
B4. One-time 30 percent real depreciation in 2010	174	181	188	173	161	148	109	94
B5. 10 percent of GDP increase in other debt-creating flows in 2010	174	218	229	214	204	191	148	123
Debt Service-to-Revenue Ratio 2/								
Baseline	13	7	8	14	16	18	20	16
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	13	7	8	14	13	14	15	2
A2. Primary balance is unchanged from 2009	13	7	8	15	14	15	15	5
A3. Permanently lower GDP growth 1/	13	7	8	15	17	19	26	36
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2010–11	13	7	9	18	22	26	35	41
B2. Primary balance is at historical average minus one standard deviations in 2010–11	13	7	9	28	27	26	24	21
B3. Combination of B1-B2 using one half standard deviation shocks	13	7	9	21	21	26	31	35
B4. One-time 30 percent real depreciation in 2010	13	7	9	17	19	22	25	21
B5. 10 percent of GDP increase in other debt-creating flows in 2010	13	7	12	51	20	39	25	23

Sources: C.A.R. authorities; and IMF and World Bank staff estimates and projections.
1/ Assumes that real GDP growth is at baseline minus one standard deviation divided by the square root of the length of the projection period.
2/ Revenues are defined inclusive of grants.